

Hotel Booking Demand Analytics Platform

Executive Business Report

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This report summarizes the analytical findings obtained from the Hotel Booking Demand dataset. The analysis includes booking behaviour, customer insights, revenue performance, cancellation analysis and statistical validation.

Executive Summary

Metric	Value
Total Bookings	87225
Cancellation Rate	27.52%
Average ADR	105.52
Average Stay Duration	3.58

Executive Insights

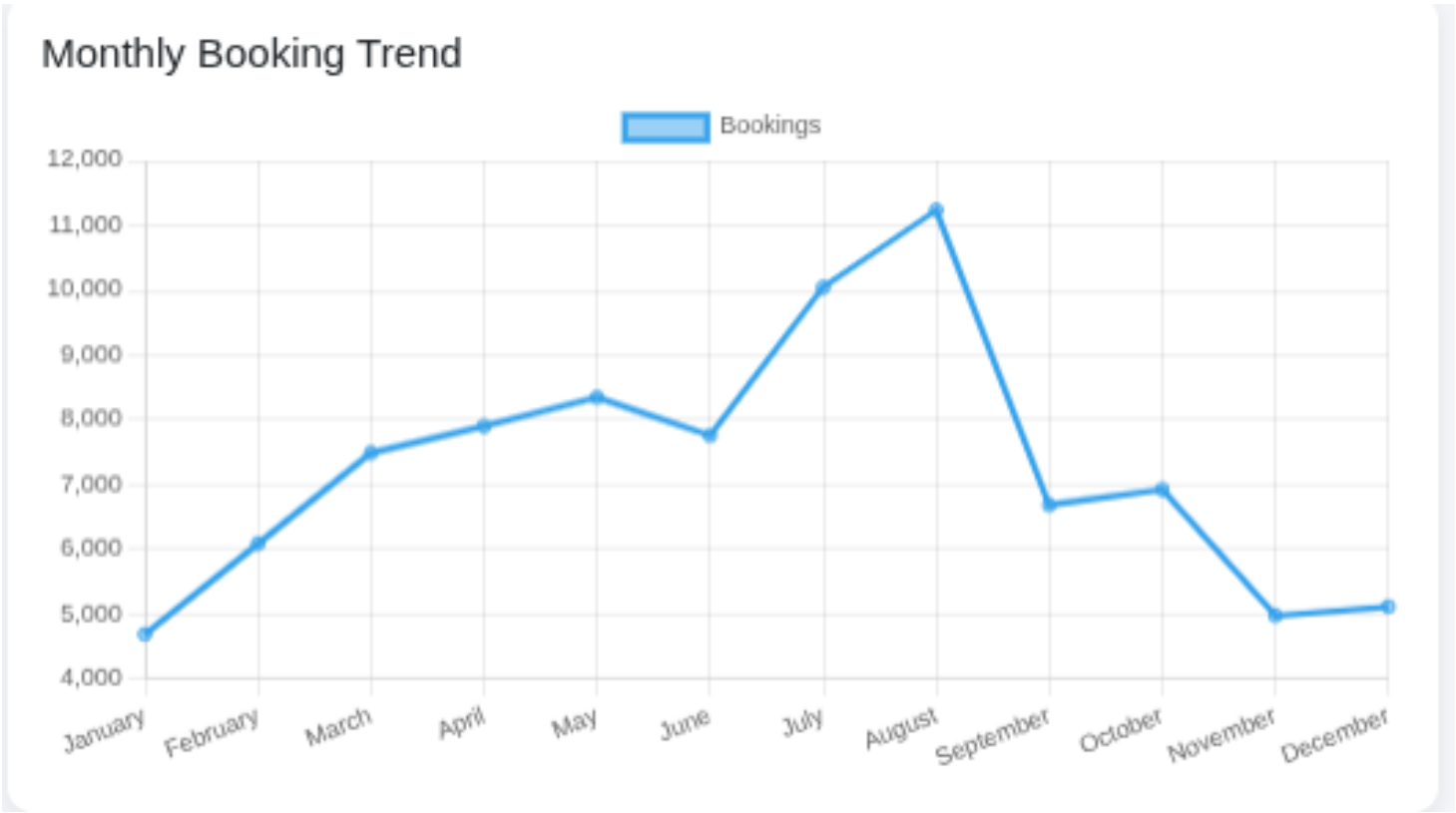
- The dataset contains **87225** hotel bookings collected from multiple countries and hotel types.
- The overall cancellation rate is **27.52%**, indicating that nearly one out of every four bookings is cancelled.
- The average Average Daily Rate (ADR) is **105.52**, which reflects the average revenue generated per occupied room.
- Guests stay for an average of **3.58** nights per booking.

Booking Analytics

Metric	Value
Peak Booking Month	August
Most Preferred Hotel	City Hotel
Average Lead Time	78.4 Days
Average Stay	3.6 Nights

Booking Insights

- **August** records the highest number of bookings.
- **City Hotel** is the most preferred hotel category.
- Customers book hotels approximately **78.4** days before arrival.
- Average stay duration is **3.6** nights.



Customer Analytics

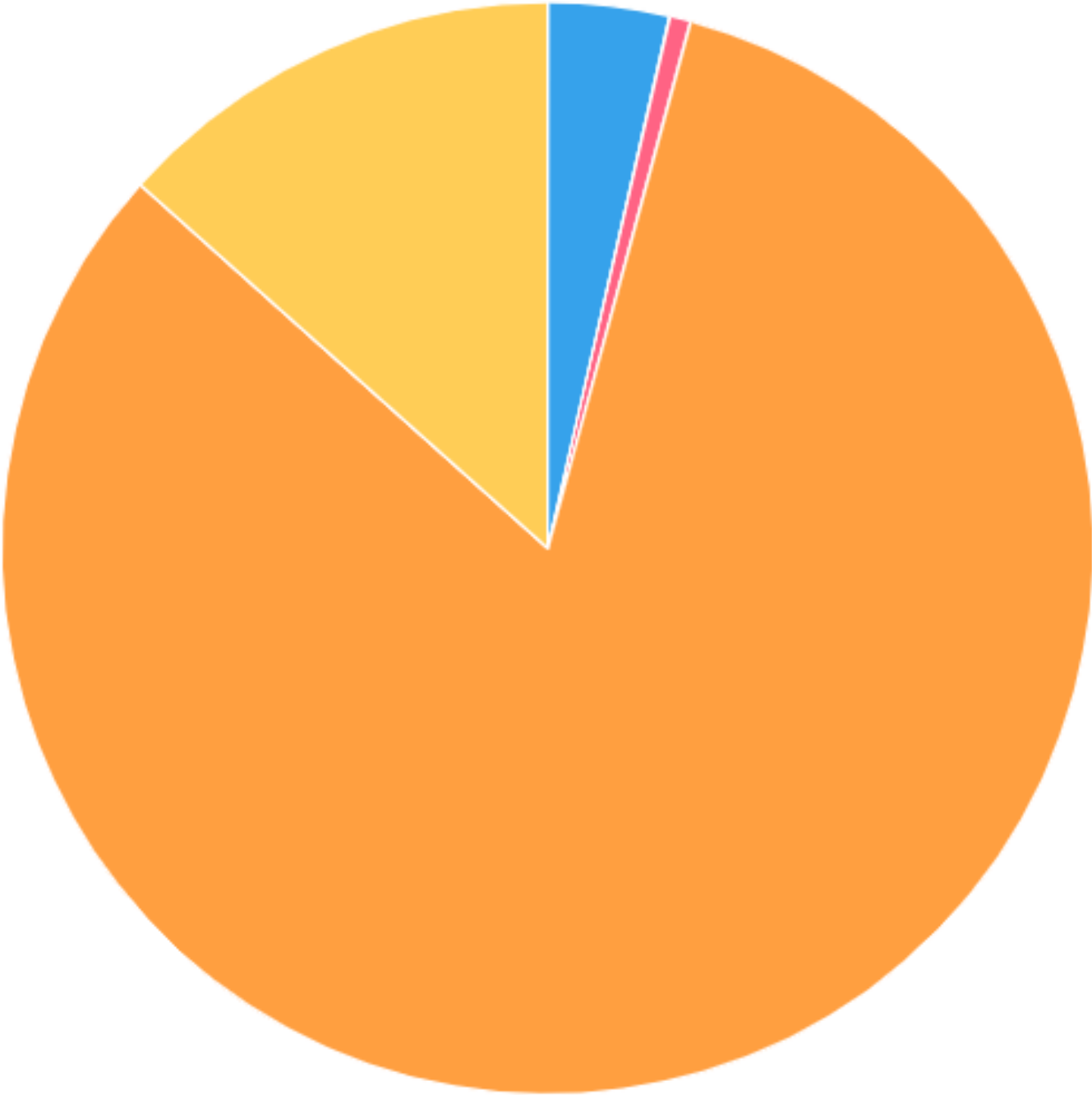
Metric	Value
Total Customers	87225
Repeat Guests	3363
New Guests	83862
Total Countries	177

Customer Insights

- Bookings were received from **177** countries.
- The majority of guests are first-time visitors.
- Only a small percentage consists of repeat guests, indicating an opportunity to improve customer loyalty.
- Customer distribution varies across market segments, helping hotels understand demand sources.

Customer Type

Contract Group Transient Transient-Party

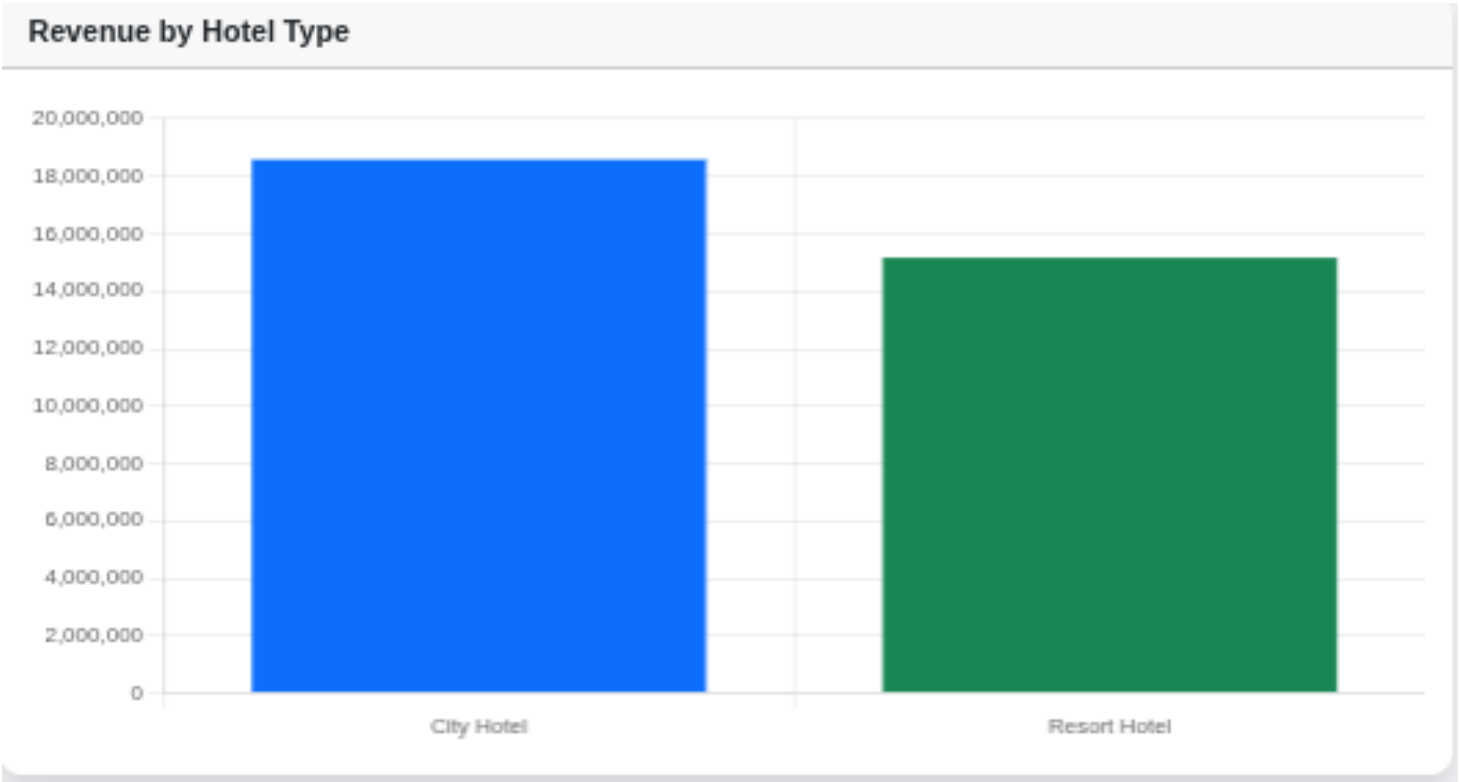


Revenue Analytics

Metric	Value
Total Revenue	33726511.74
Average ADR	105.52
Maximum ADR	226.88
Minimum ADR	0.0

Revenue Insights

- Total estimated revenue exceeds **33726511.74**.
- Average Daily Rate remains one of the strongest indicators of hotel profitability.
- Higher ADR values indicate premium pricing during high-demand periods.
- Revenue optimization can be achieved using seasonal pricing strategies.



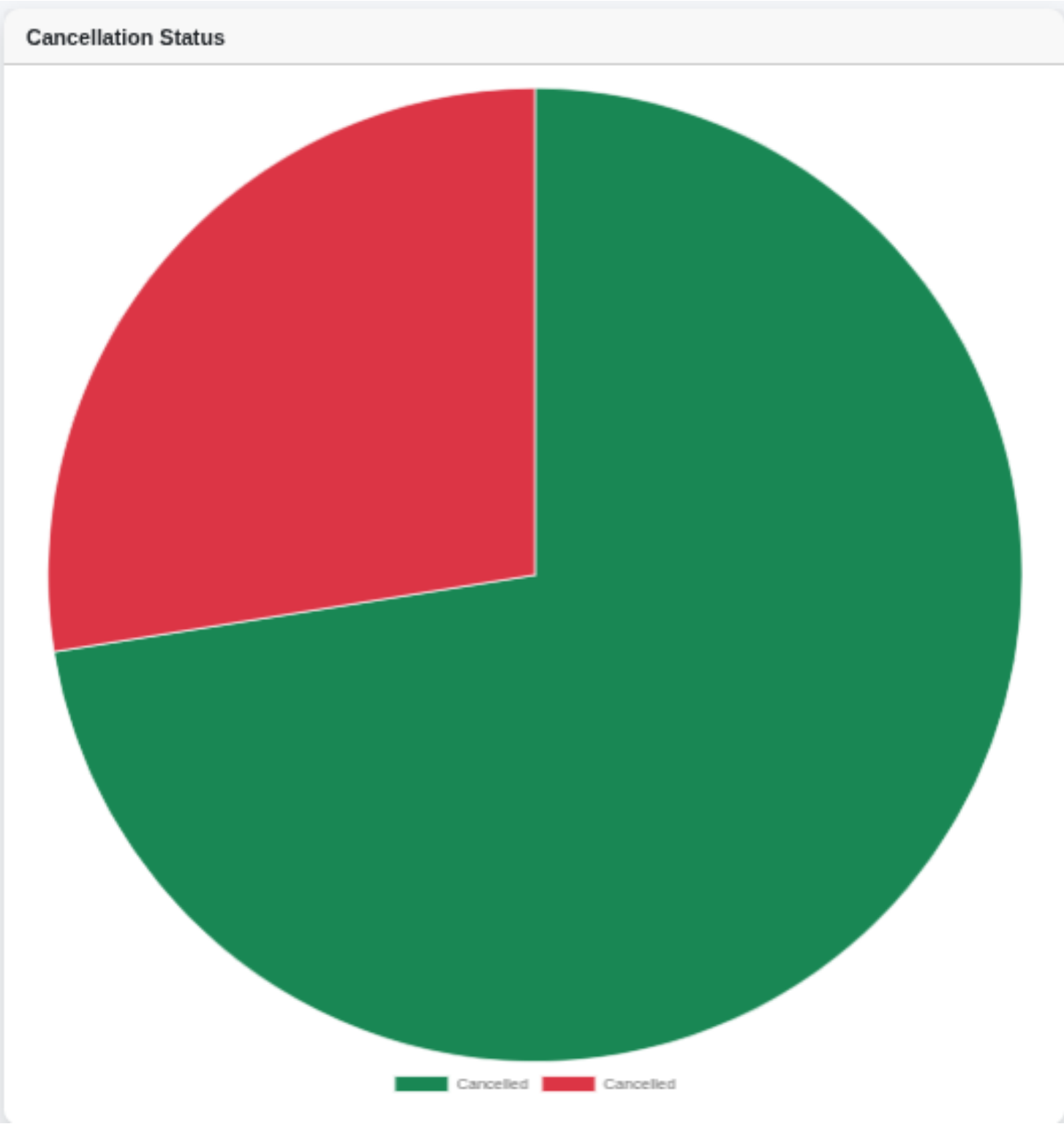
Cancellation Analytics

Metric	Value
Cancelled Bookings	24005
Non-Cancelled Bookings	63220

Highest Cancellation Segment	Online TA
Average Lead Time (Cancelled)	103.23

Cancellation Insights

- Approximately **status count 0 False 63220 1 True 24005**% of bookings were cancelled.
- Guests with longer lead times tend to cancel more frequently.
- Certain market segments contribute more heavily to cancellations.
- Understanding cancellation behaviour helps reduce revenue loss.

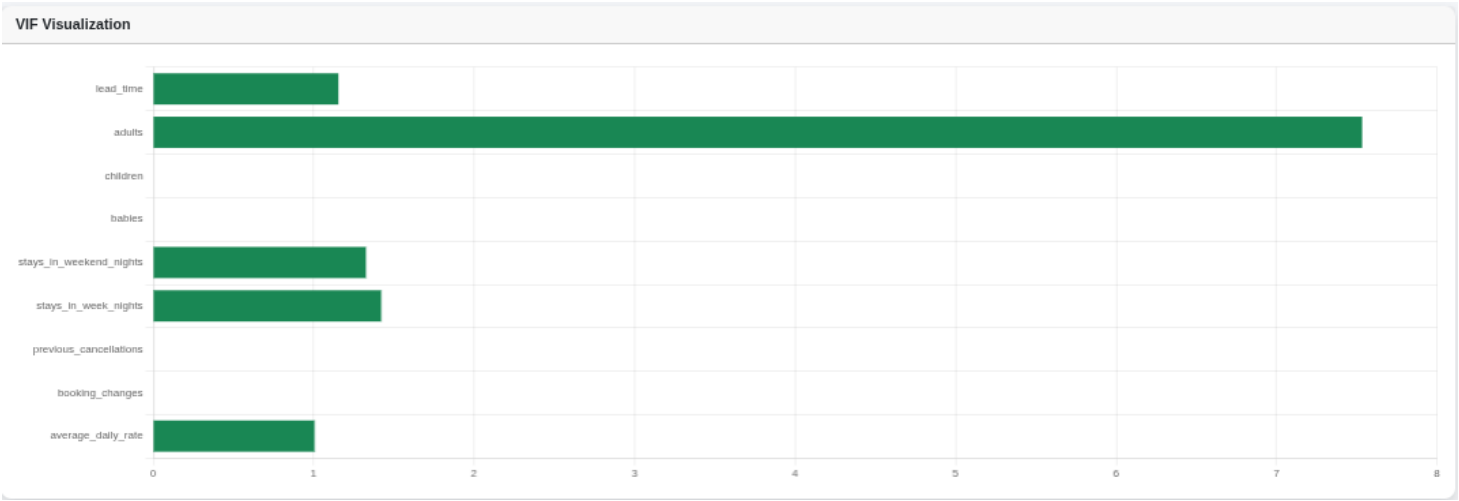


Statistical Analysis

Test	Result
Normality Test	Not Normal
Highest VIF	7.53
Strong Correlations	3
Independent t-Test	37.05
Chi-Square Test (P-value)	9.41649475121818e-297

Statistical Interpretation

- The dataset is **Not Normal** according to the normality test.
- Maximum VIF value is **7.53**, indicating acceptable multicollinearity.
- A total of **3** strong feature correlations were identified.
- The Independent t-test produced a statistic of **37.05**, confirming statistically significant differences.
- Chi-square test produced a p-value of **9.41649475121818e-297** indicating significant association between categorical variables.



Business Insights

- Peak booking demand occurs during the summer season.
- City Hotels contribute the largest share of bookings.
- Long lead-time bookings have a higher cancellation probability.
- Average Daily Rate increases significantly during peak months.
- Most bookings originate from Online Travel Agencies.

Key Business Recommendations

- Implement dynamic pricing during peak booking months.
- Focus marketing efforts on high-revenue customer segments.
- Introduce loyalty programs to increase repeat guest bookings.
- Reduce cancellations through flexible booking policies and reminder notifications.
- Use predictive analytics to forecast occupancy and optimize room pricing.

Final Conclusion

The Hotel Booking Demand Analytics Platform successfully analyzed booking patterns, customer behaviour, revenue performance, cancellation trends and statistical characteristics of the dataset. The project combines exploratory data analysis, statistical testing and interactive dashboards to support data-driven business decision making.
